

FILE PERMISSIONS

How to give read and write permissions using Unix commands. A very simple example is to create a text file with a C program, then use `chmod`.

Create a file called `permission.c`

```
#include <stdio.h> int
main()
{
    FILE *fp;    fp =
fopen("sample.txt", "w");    if
(fp == NULL)
    {        printf("Cannot open
file\n");        return 1;
    }    fprintf(fp, "Hello, this is a test
file.\n");    fclose(fp);    printf("File
created successfully.\n");    return 0;
}
```

Compile `gcc permission.c -o permission`

Run

`./permission`

Output

File created successfully.

A file called sample.txt is created.

Check the current permissions

```
ls -l sample.txt
```

You may see

```
-rw-r--r-- 1 user user 31 Aug 17 08:20 sample.txt
```

The owner has: Read + Write

Group has: Read Others

have: Read

This is permission 644.

Give Read and Write permission to everyone `chmod`

```
666 sample.txt
```

Check `ls -l sample.txt`

You should see

```
-rw-rw-rw- 1 user user ... sample.txt
```

Instead of numbers, you can directly give:

```
chmod u+rw sample.txt chmod  
o+rw sample.txt chmod a+rw  
sample.txt
```

Task of the Day

- 1) Give Read and Write only to the owner

```
venu@LAPTOP-KHLR30TF:~/os-lab$ chmod 600 sample.txt  
venu@LAPTOP-KHLR30TF:~/os-lab$ ls -l sample.txt  
-rw----- 1 venu venu 28 Aug 24 05:38 sample.txt  
venu@LAPTOP-KHLR30TF:~/os-lab$ |
```

- 2) Give Read and Write to the owner and group

```
venu@LAPTOP-KHLR30TF:~/os-lab$ chmod 660 sample.txt  
venu@LAPTOP-KHLR30TF:~/os-lab$ ls -l sample.txt  
-rw-rw---- 1 venu venu 28 Aug 24 05:38 sample.txt  
venu@LAPTOP-KHLR30TF:~/os-lab$ |
```

- 3) Only owner can read and write

```
venu@LAPTOP-KHLR30TF:~/os-lab$ chmod 600 sample.txt  
venu@LAPTOP-KHLR30TF:~/os-lab$ ls -l sample.txt  
-rw----- 1 venu venu 28 Aug 24 05:38 sample.txt  
venu@LAPTOP-KHLR30TF:~/os-lab$ |
```

- 4) Owner read/write, everyone else read-only

```
venu@LAPTOP-KHLR30TF:~/os-lab$ chmod 644 sample.txt  
venu@LAPTOP-KHLR30TF:~/os-lab$ ls -l sample.txt  
-rw-r--r-- 1 venu venu 28 Aug 24 05:38 sample.txt  
venu@LAPTOP-KHLR30TF:~/os-lab$ |
```

- 5) Owner can do everything, others can only read and execute

```
venu@LAPTOP-KHLR30TF:~/os-lab$ chmod 755 sample.txt
venu@LAPTOP-KHLR30TF:~/os-lab$ ls -l sample.txt
-rwxr-xr-x 1 venu venu 28 Aug 24 05:38 sample.txt
venu@LAPTOP-KHLR30TF:~/os-lab$ |
```

- 6) Owner and group can read/write

```
venu@LAPTOP-KHLR30TF:~/os-lab$ chmod 660 sample.txt
venu@LAPTOP-KHLR30TF:~/os-lab$ ls -l sample.txt
-rw-rw---- 1 venu venu 28 Aug 24 05:38 sample.txt
venu@LAPTOP-KHLR30TF:~/os-lab$ |
```

- 7) Only owner can execute

```
venu@LAPTOP-KHLR30TF:~/os-lab$ chmod 700 sample.txt
venu@LAPTOP-KHLR30TF:~/os-lab$ ls -l sample.txt
-rwx----- 1 venu venu 28 Aug 24 05:38 sample.txt
venu@LAPTOP-KHLR30TF:~/os-lab$ |
```

Fill in the permission numbers for the following:

- 8) Owner: read + write; Group: nothing; Others: nothing
→ 600
- 9) Owner: read + write; Group: read; Others: read →
644
- 10) Owner: read + write + execute; Group: read + execute;
Others: read + execute → 755
- 11) Owner: read + write; Group: read + write; Others:
nothing → 660
- 12) Everyone: read only → 444

```
venu@LAPTOP-KHLR30TF:~$ cd os-lab
venu@LAPTOP-KHLR30TF:~/os-lab$ nano permission.c
venu@LAPTOP-KHLR30TF:~/os-lab$ gcc permission.c -o permission
venu@LAPTOP-KHLR30TF:~/os-lab$ ./permission
File created successfully.
venu@LAPTOP-KHLR30TF:~/os-lab$ chmod 666 sample.txt
venu@LAPTOP-KHLR30TF:~/os-lab$ ls -l sample.txt
-rw-rw-rw- 1 venu venu 28 Aug 24 05:38 sample.txt
venu@LAPTOP-KHLR30TF:~/os-lab$ chmod 600 sample.txt
venu@LAPTOP-KHLR30TF:~/os-lab$ ls -l sample.txt
-rw----- 1 venu venu 28 Aug 24 05:38 sample.txt
venu@LAPTOP-KHLR30TF:~/os-lab$ chmod 660 sample.txt
venu@LAPTOP-KHLR30TF:~/os-lab$ ls -l sample.txt
-rw-rw---- 1 venu venu 28 Aug 24 05:38 sample.txt
venu@LAPTOP-KHLR30TF:~/os-lab$ chmod 600 sample.txt
venu@LAPTOP-KHLR30TF:~/os-lab$ ls -l sample.txt
-rw----- 1 venu venu 28 Aug 24 05:38 sample.txt
venu@LAPTOP-KHLR30TF:~/os-lab$ chmod 644 sample.txt
venu@LAPTOP-KHLR30TF:~/os-lab$ ls -l sample.txt
-rw-r--r-- 1 venu venu 28 Aug 24 05:38 sample.txt
venu@LAPTOP-KHLR30TF:~/os-lab$ chmod 755 sample.txt
venu@LAPTOP-KHLR30TF:~/os-lab$ ls -l sample.txt
-rwxr-xr-x 1 venu venu 28 Aug 24 05:38 sample.txt
venu@LAPTOP-KHLR30TF:~/os-lab$ chmod 660 sample.txt
venu@LAPTOP-KHLR30TF:~/os-lab$ ls -l sample.txt
-rw-rw---- 1 venu venu 28 Aug 24 05:38 sample.txt
venu@LAPTOP-KHLR30TF:~/os-lab$ chmod 700 sample.txt
venu@LAPTOP-KHLR30TF:~/os-lab$ ls -l sample.txt
-rwx----- 1 venu venu 28 Aug 24 05:38 sample.txt
venu@LAPTOP-KHLR30TF:~/os-lab$ |
```